

```
1 void spmv(v_ty* x, const csr_ty* a, const v_ty* b) {
2     v_ty sum = 0;
3     for (i_ty i = 0; i < a->num_rows; i++) {
4         p_ty p_beg = a->ptrs[i], p_end = a->ptrs[i+1];
5         for (p_ty p = p_beg; p < p_end; p++) {
6             v_ty vec_val = b[a->idxs[p]];
7             v_ty nnz_val = a->vals[p];
8             sum += vec_val * nnz_val; //inner-loop body (ri)
9         }
10        x[i] = sum; sum = 0; //inner-loop tail (re)
11    }
12 }
```